

EXPERIMENT NO.4

Aim: To Design and develop a program for Crypto currency using MetaMask.

Theory:

What is a cryptocurrency - Crypto (short for cryptocurrency) is a new form of digital money powered by cryptography. It all started in 2008 with Bitcoin. You could use it to send funds to anyone anywhere globally. What made crypto different from normal bank transfers or other financial services like PayPal or Alipay is that there was no middleman for the first time.

Things are different with crypto. **Transactions directly connect sender and recipient** without having to deal with any central authority. Nobody else will have access to your funds and **nobody can tell you what services you can use**. This is possible because of the blockchain technology upon which cryptocurrencies operate.

1. Ethereum [ETC] -

Ethereum is a decentralized, open-source blockchain with smart contract functionality. Ether is the native cryptocurrency of the platform. Among cryptocurrencies, ether is second only to bitcoin in market capitalization. **Launched in 2015, Ethereum builds on Bitcoin's innovation, with some big differences.**

Ethereum is a blockchain-based distributed platform. The network currency of Ethereum is known as Ether (ETH). Here also, the transactions are stored in an immutable distributed ledger

- Ethereum is designed to be scalable, decentralized, and programmable.
- It provides a flexible platform to build applications using the solidity scripting language.
- Transactions are sent and received in user-created Ethereum accounts.
- It is a blockchain-based platform with the cryptocurrency Ether (ETH).

2. Bitcoin [BTC] –

Bitcoin is a digital currency that can be transferred on a peer-to-peer (P2P) network without the need for any central authority. It was invented by a person or group of people with the name Satoshi Nakamoto in 2008. All the transactions are stored in an immutable distributed ledger.

- Bitcoin is created, stored, transacted, and distributed using a decentralized, distributed system known as Blockchain.
- A public ledger records all the transactions of the Bitcoin and copies are retained on all the servers around the world.
- It is not necessary to buy an entire bitcoin, one can buy only a fraction of it if that is all necessary.
- BTC is the abbreviation for Bitcoin.

Difference between Ethereum and Bitcoin –

Basis	Bitcoin	Ethereum
Definition	Bitcoin (abbreviation: BTC; sign: ₿) is a decentralized digital currency that can be transferred on the peer-to-peer bitcoin network.	Ethereum is a decentralized global software platform powered by blockchain technology. It is most commonly known for its native cryptocurrency, ether (ETH).
History	The word bitcoin was defined in a white paper published on 31 October 2008. The currency began use in 2009.	Ethereum was conceived in 2013 by programmer Vitalik Buterin, and then went live on 30 July 2015.
Purpose	The purpose of bitcoin was to replace national currencies during the financial crisis of 2008.	The purpose of Ethereum was to utilize blockchain technology for maintaining a decentralized payment network and storing computer code.
Smart Contracts	Although bitcoin do have smart contracts, they are not as flexible or complete as Ethereum smart contracts. Smart contracts in Bitcoin does not have all the functionality that a programming language would give them.	Ethereum allows us to create smart contracts. Smart contracts are computer codes that is stored on a blockchain and executed when the predetermined terms and conditions are met.

Smart Contract Programming Language	Smart contracts on Bitcoin are written in programming languages like Script, Clarity.	Smart contracts on Ethereum are written in programming languages like Solidity, Vyper, etc.
Transactions	Generally, bitcoin transactions are only for keeping notes.	Ethereum transactions may contain some executable code.
Hash Algorithm	Bitcoin runs on the SHA-256 hash algorithm.	Ethereum runs on the Keccak-256 hash algorithm.
Consensus Mechanism	The Proof-of-Work (PoW) is the consensus mechanism used by the Bitcoin network.	The Proof-of-Stake is the consensus mechanism used by Ethereum.
Block Time	The block time of bitcoin is 10 minutes.	The block time of Ethereum is 14 to 15 seconds.
Block Limit	The bitcoin blockchain has a block limit of 1 MB.	The Ethereum blockchain does not have a block limit.
Popularity	Bitcoin is the most popular digital currency in the market to date.	Ether, native currency of Ethereum, is the second-Largest cryptocurrency after bitcoin to date.

What is MetaMask –

Founded in 2016 by blockchain software company ConsenSys, MetaMask is a crypto wallet that supports Ether and other ERC-20 tokens. You can use it from your PC or laptop as a browser extension or install its mobile application on your smartphone.

MetaMask is one of the most popular browser extensions that serves as a way of storing your Ethereum and other ERC-20 Tokens. The extension is free and secure, allowing web applications to read and interact with Ethereum's blockchain.

With MetaMask, you can access all the decentralised applications of the Ethereum blockchain, invest in NFTs and utilise smart contracts. Moreover, as it is a browser extension on your computer, it saves a lot of space since you do not have to sync the entire blockchain in your system.

MetaMask is a **popular cryptocurrency wallet** that supports a broad range of Ethereum-based tokens and non-fungible tokens (NFTs) on supported blockchains. While experienced crypto investors may appreciate the speed and simplicity of the wallet, new investors may find it difficult to navigate.

Benefits of using MetaMask wallet

Its massive global user and follower base make it a vital part of the Ethereum community.

Utilizes local key storage, giving users full control over their keys. No personal information is stored either.

Allows in-app coin purchasing – users can buy Ether and ERC-20 directly from Coinbase and ShapeShift.

Allows multiple accounts, which is great for people who want to keep their finances separate.

Backs up the user account with hierarchical deterministic settings.

Has an easy-to-use interface that is good for crypto beginners.

Easily connects with popular exchanges like Coinbase and Binance.



| cointelegraph.com

To Download and install MetaMask, follow the given steps:

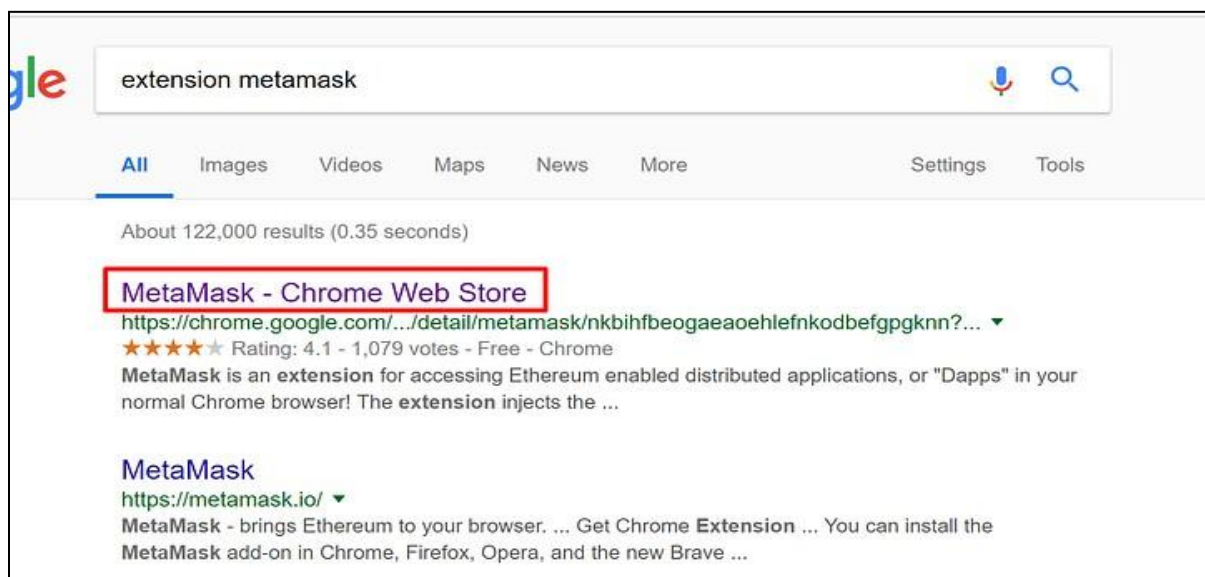
1. Go to the MetaMask homepage, press 'Download', and click on 'Install MetaMask for Chrome'. You will now be redirected to the Chrome web store.

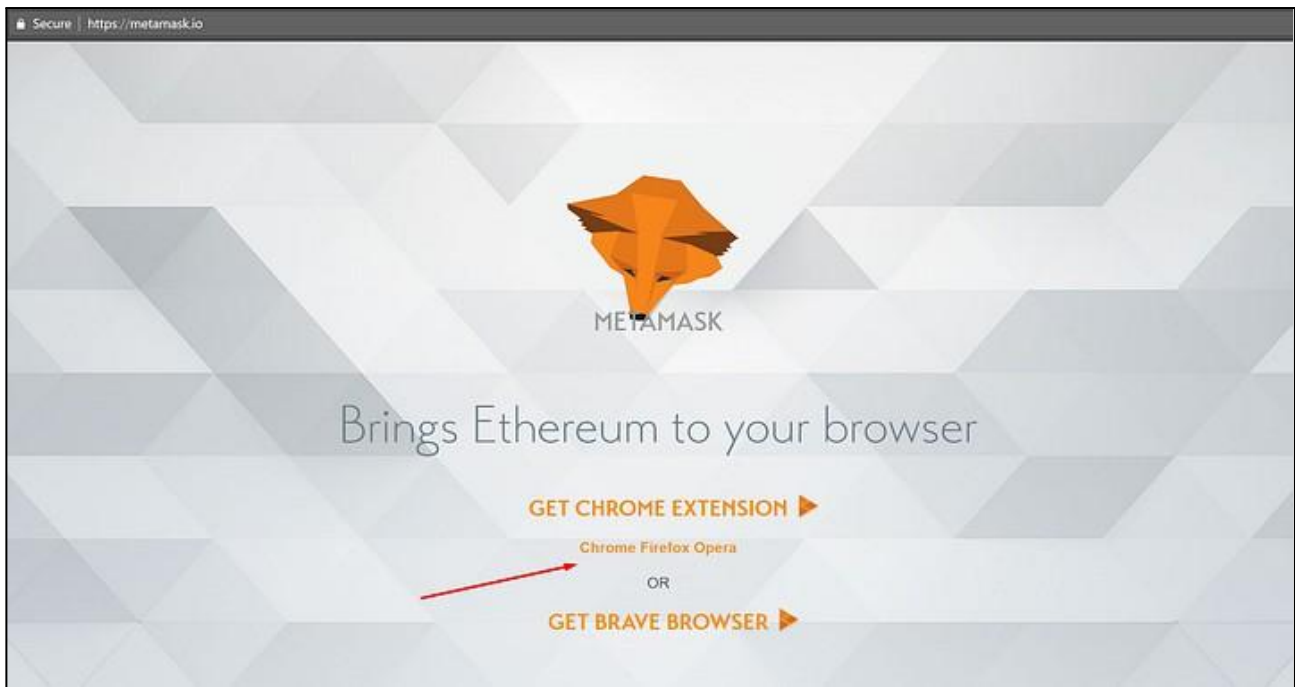


- Install MetaMask on your browser.

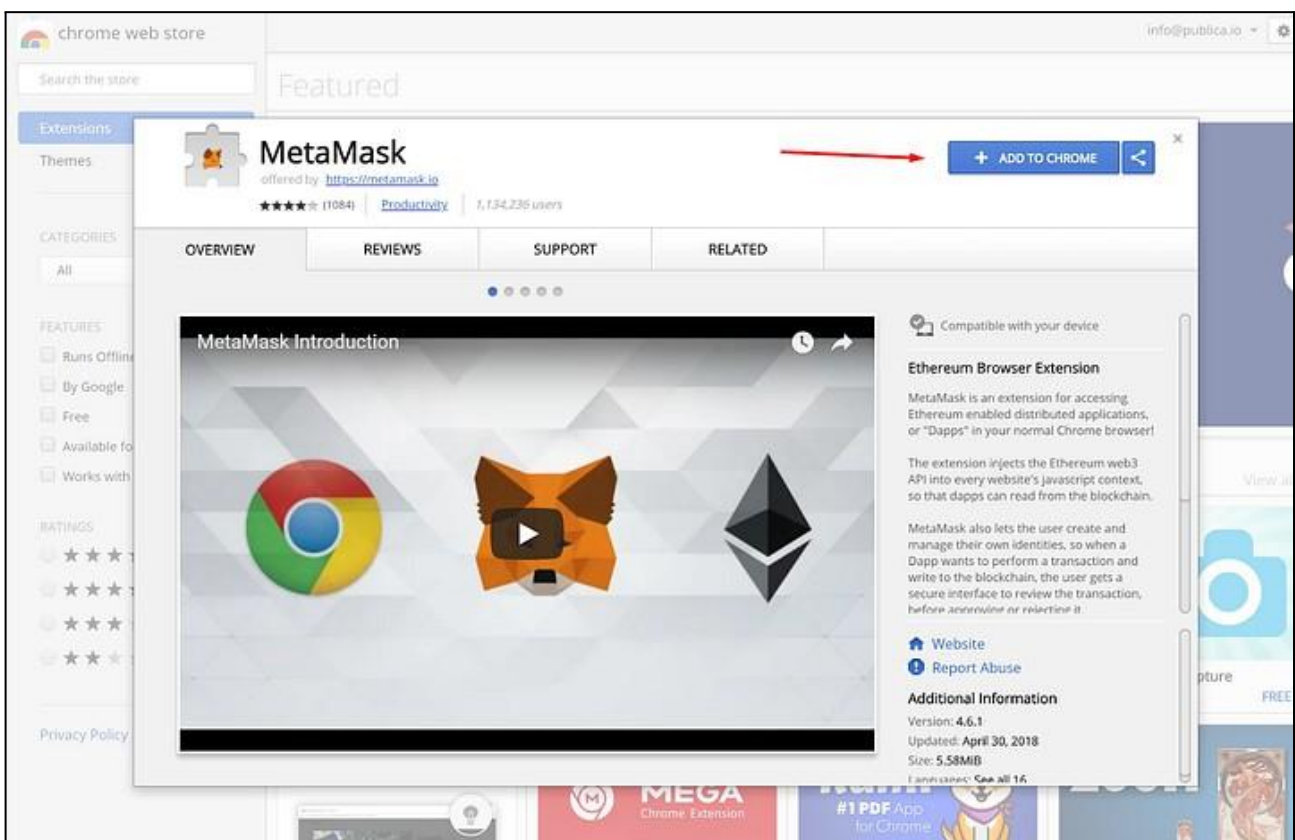
To create a new wallet, you have to install the extension first. Depending on your browser, there are different marketplaces to find it. Most browsers have MetaMask on their stores, so it's not that hard to see it, but either way, here they are Chrome, Firefox, and Opera.

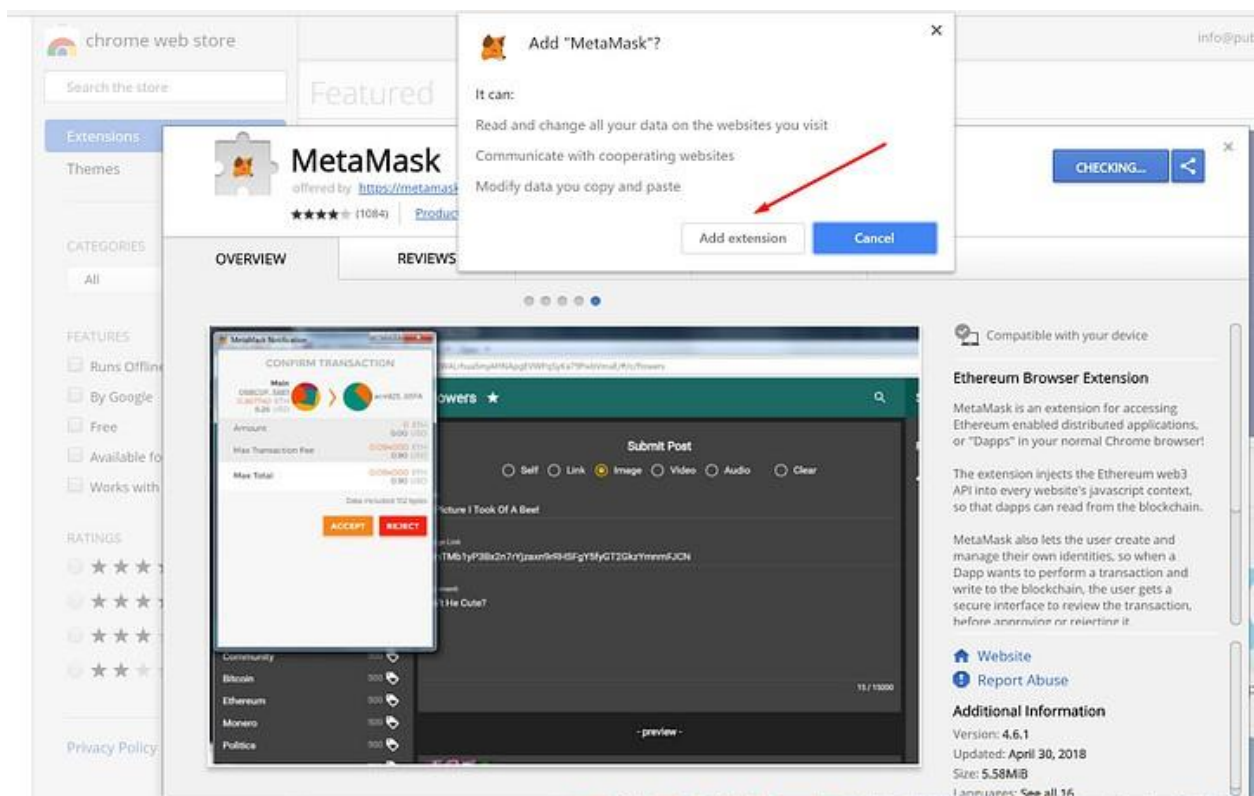
We will be using Google Chrome as an example but the same steps apply to different browsers as well. If, for some reason, the links above don't work for you, just search MetaMask Extension on your favourite search engine, and you can find it there.





- Click on Install MetaMask as a Google Chrome extension.
- Click Add to Chrome → Click Add Extension.





And it's as easy as that to install the extension on your browser, continue reading the next step to figure out how to create an account.

2. Create an account.

- Click on the extension icon in the upper right corner to open MetaMask.
- To install the latest version and be up to date, click Try it now.
- You will be prompted to create a new password. Click Create.

MetaMask
chrome-extension://nkbihfbeogaeaoehlefnkodbefgpgknn/home.html#initialize/create-password

Create Password

New Password (min 8 chars)

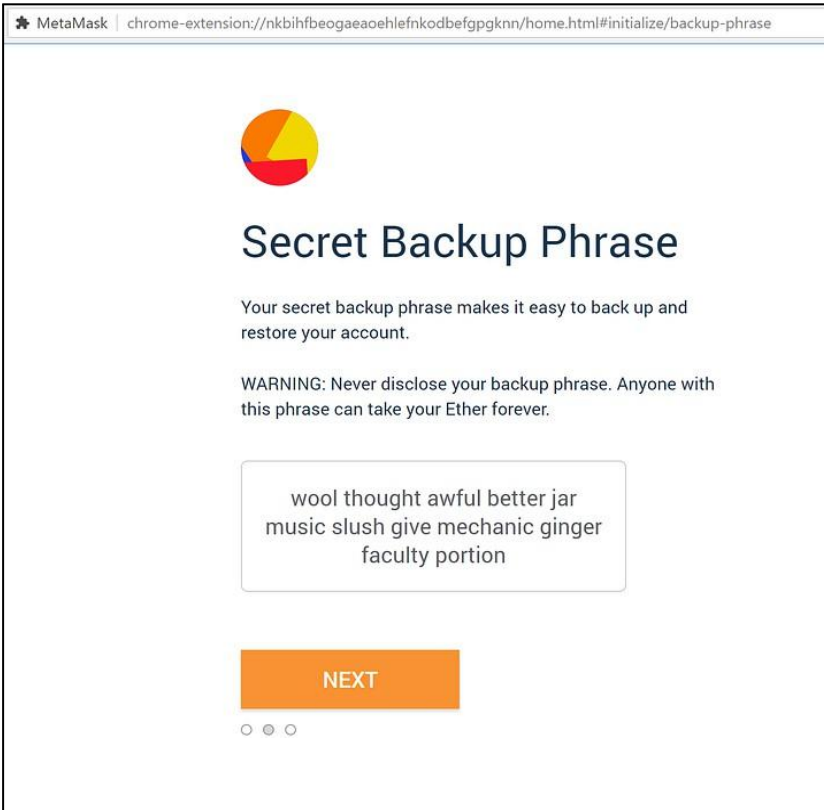
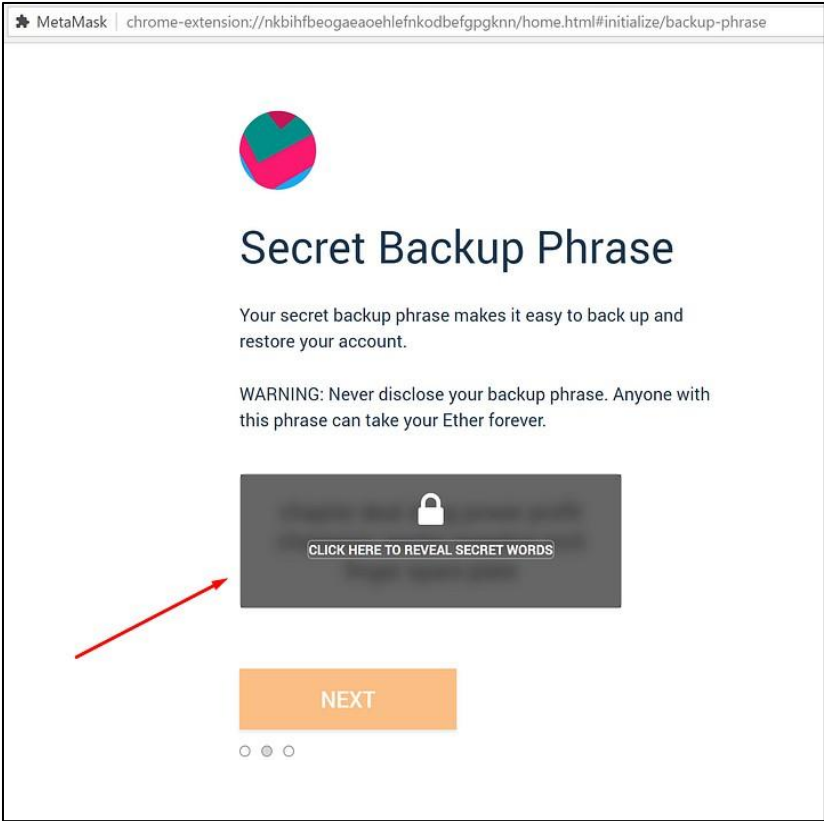
Confirm Password

CREATE

Import with seed phrase

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Click Reveal Secret Words. There you will see 12 words seed phrase. This is really important and usually not a good idea to store digitally, so take your time and write it down.

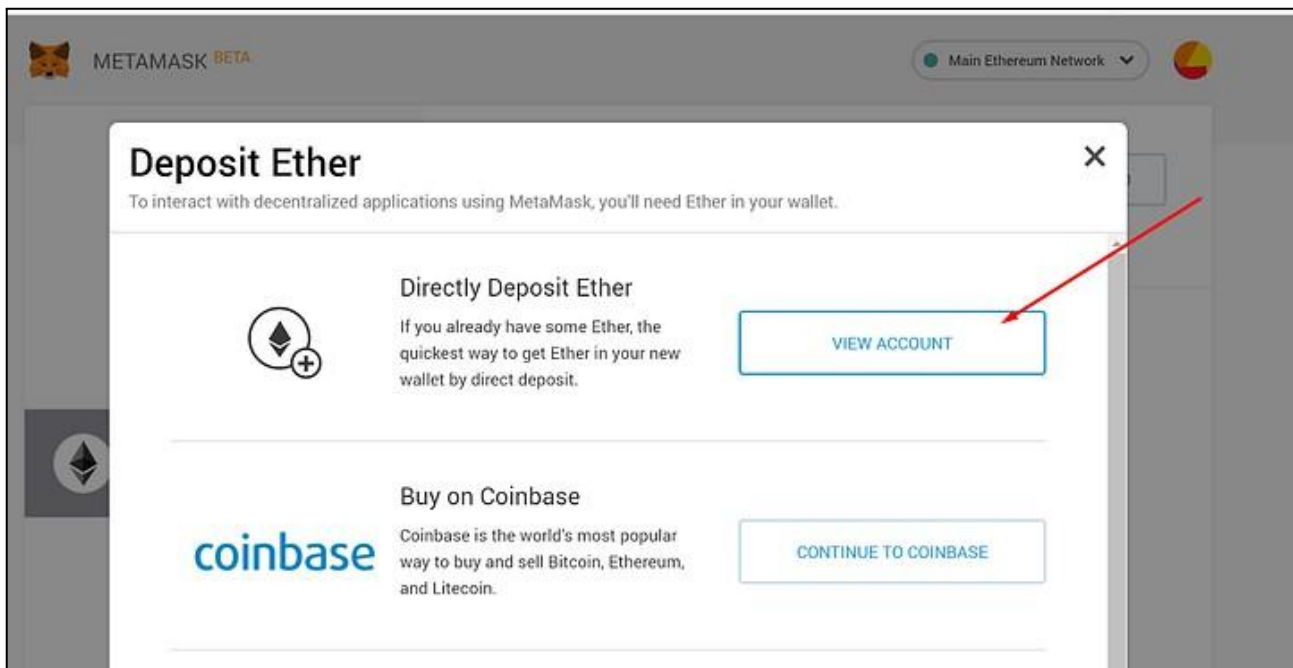


- Verify your secret phrase by selecting the previously generated phrase in order. Click Confirm.

And that's it; now you have created your MetaMask account successfully. A new Ethereum wallet address has just been created for you. It's waiting for you to deposit funds, and if you want to learn how to do that, look at the next step below.

3. Depositing funds.

- Click on View Account.
- You can now see your public address and share it with other people. There are some methods to buy coins offered by MetaMask, but you can do it differently as well; you just need your address.



The MetaMask extension will always be available in your browser's toolbar, ready for you to continue the journey into the Crypto world. It's easily accessible and, as you saw from these steps, simple to use as well. Sending coins is also very intuitive; there's a big send button, and the rest are easily understandable. But in case something wrong happens, and you need to restore your account, you have this option. First, you have to back it up, and let's look at how we can do that.

Conclusion: Thus, we have successfully studied and developed a program for Crypto currency using MetaMask.